

6. $f(x) = \begin{cases} 0 & \text{for } x < 0 \\ x^2 & \text{for } x \geq 0 \end{cases}$ is monotonically increasing on $\mathbb{R}$, and it is *strictly* increasing on $[0, +\infty)$.

Proposition 4.1: A function that is strictly increasing/decreasing on its domain is injective (one-to-one).

Proof. Consider first the case that $f : \text{dom}(f) \subseteq \mathbb{R} \rightarrow \mathbb{R}$ is strictly increasing on $\text{dom}(f)$. Let $x_1, x_2 \in \text{dom}(f)$ with $x_1 \neq x_2$. Without loss of generality $x_1 < x_2$, so that $f(x_1) < f(x_2)$. In particular, $f(x_1) \neq f(x_2)$, so that f is injective. The case of a strictly decreasing function follows the same idea of proof. $\square$

★★★ The converse statement – i.e. that an injective function is strictly monotone – is not true, see Figure 4.6 for a counterexample.

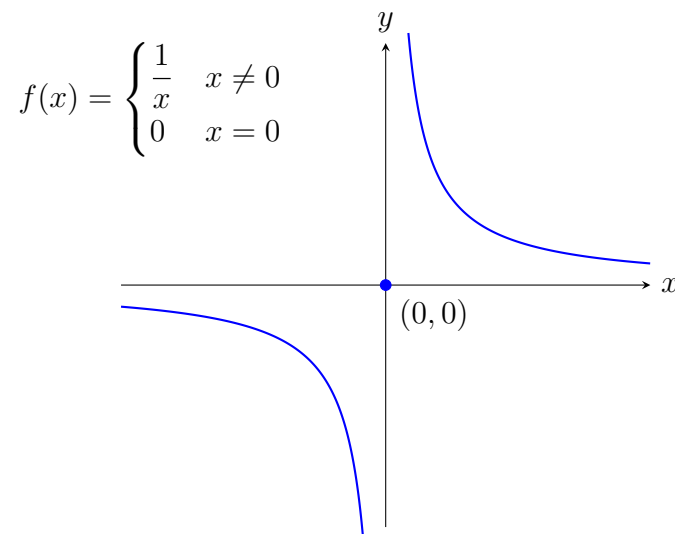


Figure 4.6: A one-to-one function that is neither increasing nor decreasing

Lemma 4.2: Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be monotonically increasing on some $A \subseteq \mathbb{R}$. Then $f + g$ is also monotonically increasing on A . If either f or g are *strictly* increasing on A , then so is $f + g$. The same statements hold if we replace everywhere the word ‘increasing’ with the word ‘decreasing’.

Proof. Exercise. $\square$

4.5 Composition of functions

The composition of functions – i.e. the application of two (or more) functions successively – is something that often comes up in mathematics and its applications.

Example 4.5 (Taxi fare): Suppose that the fare for riding a taxi is made of a flat fee of 3 Euros plus twice the distance travelled (in kilometers). So the fee for riding x kilometers is:

$$f(x) = 2x + 3.$$

Now, suppose that a card payment carries a 5% surcharge of the total fare:

$$g(y) = 1.05y.$$

So, if we travel x kilometers, and want to pay by card, the total amount to pay is:

$$g(f(x)) = 1.05(2x + 3) = 2.1x + 3.15$$

This is a composition of functions.

Let X, Y, Z be sets and let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ be two functions. The **composition of f and g** is a new function $h : X \rightarrow Z$ defined as

$$h(x) = g(f(x)).$$

It is denoted by $h = g \circ f$ so we can also write $h(x) = (g \circ f)(x)$.

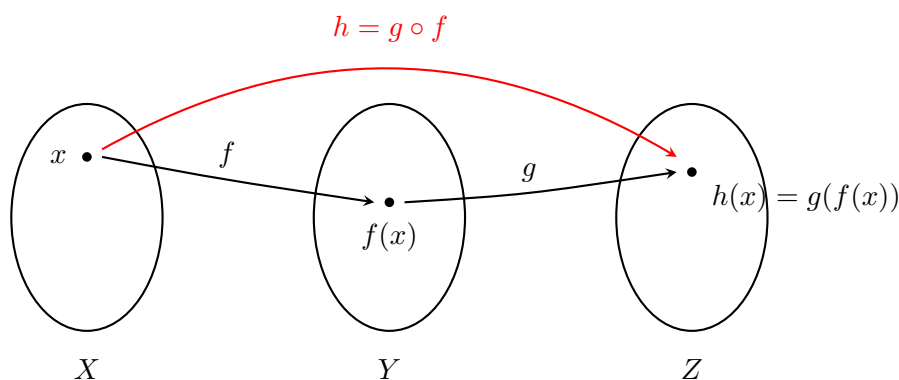


Figure 4.7: Composition of functions $h(x) = g(f(x))$

The domain of h is defined as follows:

$$x \in \text{dom}(h) \quad \Leftrightarrow \quad x \in \text{dom}(f) \quad \text{and} \quad f(x) \in \text{dom}(g).$$

Example 4.6: 1. If $f(x) = x^2$, $g(y) = y - 3$, then $h(x) = g(f(x)) = x^2 - 3$, and $\text{dom}(h) = \text{dom}(f)$.

2. If $f(x) = e^x$, $g(y) = -y$, then $h(x) = g(f(x)) = -e^x$, and $\text{dom}(h) = \text{dom}(f)$.

3. If $f(x) = -x$, $g(y) = e^y$, then $h(x) = g(f(x)) = e^{-x}$, and $\text{dom}(h) = \text{dom}(f)$.

4. If $f(x) = \sqrt{x}$, $g(y) = y^2$, then $h(x) = g(f(x)) = (\sqrt{x})^2 = x$, and $\text{dom}(h) = \text{dom}(f) = [0, +\infty)$.

5. If $f(x) = x^2$, $g(y) = \sqrt{y}$, then $h(x) = g(f(x)) = \sqrt{x^2} = |x|$, and $\text{dom}(h) = \text{dom}(f) = \mathbb{R}$.

6. If $f(x) = \frac{1}{x}$, $g(y) = \sin y$, then $h(x) = g(f(x)) = \sin \frac{1}{x}$, and $\text{dom}(h) = \text{dom}(f) = \mathbb{R} \setminus \{0\}$.

7. If $f(x) = \sin x$, $g(y) = \frac{1}{y}$, then $h(x) = g(f(x)) = \frac{1}{\sin x}$, and $\text{dom}(h) \neq \text{dom}(f)$. In this case $\text{dom}(h)$ is all $x \in \mathbb{R}$ s.t. $\sin x \neq 0$.

These examples show us that the **composition of functions need not be commutative**. In general, unless f and g have a special form, their order is important:

$$f \circ g \neq g \circ f.$$

We can also see that a function and its inverse ‘cancel’ one another. More precisely, if f is one-to-one (and therefore f^{-1} exists) then

$$f \circ f^{-1} = \text{Id}_{\text{dom}(f^{-1})} = \text{Id}_{\text{im}(f)} \quad \text{and} \quad f^{-1} \circ f = \text{Id}_{\text{dom}(f)}$$

where the function Id is defined as follows:

The identity function

Id is the identity function that can be defined on any subset:

$$\text{Id}_A(x) = x, \quad \forall x \in A.$$

Here A could be a subset of any set.

Proposition 4.3: Let $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ both be one-to-one functions. Then

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}.$$

Proof. First, we show that $g \circ f$ is one-to-one, so its inverse exists. Let $x_1, x_2 \in X$ with $(g \circ f)(x_1) = (g \circ f)(x_2)$. Then $g(f(x_1)) = g(f(x_2))$. Since g is one-to-one, $f(x_1) = f(x_2)$. Since f is one-to-one, $x_1 = x_2$. Thus $g \circ f$ is one-to-one.

Now we show $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. Let $z \in \text{dom}((g \circ f)^{-1}) = \text{im}(g \circ f)$. Then there exists a unique $x \in X$ such that $(g \circ f)(x) = z$, and $(g \circ f)^{-1}(z) = x$. We have $g(f(x)) = z$, so $f(x) = g^{-1}(z)$, and thus $x = f^{-1}(g^{-1}(z)) = (f^{-1} \circ g^{-1})(z)$. Therefore $(g \circ f)^{-1}(z) = (f^{-1} \circ g^{-1})(z)$ for all $z \in \text{dom}((g \circ f)^{-1})$. $\square$

Lemma 4.4: Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$. Then:

$$\begin{array}{lll} f, g \text{ are both monotone increasing} & \Rightarrow & g \circ f \text{ is monotone increasing.} \\ f, g \text{ are both monotone decreasing} & \Rightarrow & g \circ f \text{ is monotone increasing.} \\ f, g \text{ are monotone of different kinds} & \Rightarrow & g \circ f \text{ is monotone decreasing.} \end{array}$$

Proof. Exercise. $\square$

Translations, rescalings and reflections

Here we discuss three types of simple functions that might often appear as part of a composition of functions.

Translations are an important family of functions: they simply ‘move’ the variable x by some fixed amount. Here is the simple definition. Given a fixed $c \in \mathbb{R}$, define the **translation by c** , the function $t_c : \mathbb{R} \rightarrow \mathbb{R}$, as

$$t_c(x) = x + c.$$

Here x is the variable, as always, and the number c is normally called a *parameter*. Given some function $f : \mathbb{R} \rightarrow \mathbb{R}$, we see that

$$\begin{aligned}(f \circ t_c)(x) &= f(t_c(x)) = f(x + c), \\ (t_c \circ f)(x) &= t_c(f(x)) = f(x) + c.\end{aligned}$$

The first function is a shift of the graph of f to the left by c (if $c < 0$ then the shift is to the right). The second function is a shift of the graph of f up by c (if $c < 0$ then the shift is down).

Rescalings ‘squeeze’ or ‘stretch’ a function horizontally. Here’s the definition. Given a fixed $c > 0$, define the **scaling by c** , the function $s_c : \mathbb{R} \rightarrow \mathbb{R}$, as

$$s_c(x) = cx.$$

Composition with some function $f : \mathbb{R} \rightarrow \mathbb{R}$ gives

$$(f \circ s_c)(x) = f(s_c(x)) = f(cx).$$

If $0 < c < 1$ then the graph of f is ‘stretched’ by a factor of $\frac{1}{c}$, whereas if $c > 1$ then the graph of f is ‘squeezed’ by a factor of c , see Figure 4.8.

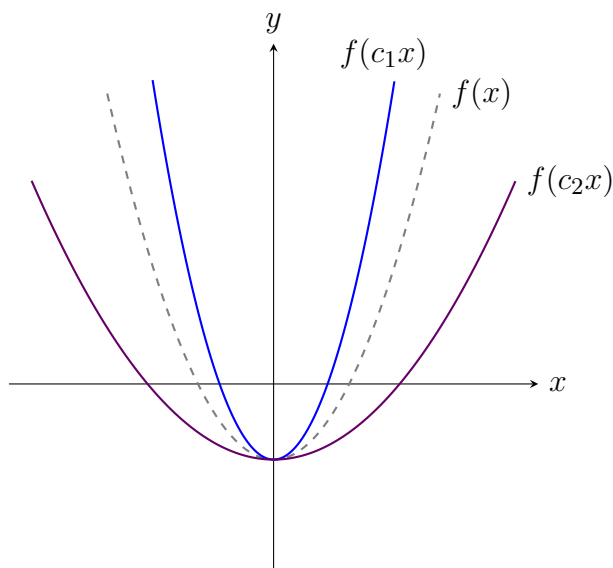


Figure 4.8: Rescalings of a function $f(x)$ by $c_1 > 1$ and $0 < c_2 < 1$.

We can also rescale a function vertically. Given a fixed $c > 0$, define

$$(s_c \circ f)(x) = s_c(f(x)) = cf(x).$$

A constant $c > 1$ will ‘stretch’ the graph of the function along the y -axis, and $0 < c < 1$ will ‘squeeze’ the graph of the function.